Aadishesh Gopal Padasalgi

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Profile — Research-driven AI/ML engineer with a strong foundation in machine learning, natural language processing, and quantum computing. Experienced in building end-to-end research systems and leading student innovation teams. Passionate about advancing interpretable and sustainable AI through academic research.

Education

B.M.S. Institute of Technology and Management — **Bengaluru, India**
B.E. in Artificial Intelligence and Machine Learning — *2023 - 2027*
Core Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Research Methodology, Data Structures and Algorithms, Artificial Intelligence, Database Management Systems, Operating Systems

Technical Skills

Languages Python, C
AI & ML TensorFlow, HuggingFace, Ollama
Quantum Qiskit, PennyLane, lambeq
Dev Tools Git, Linux, VS Code, Jupyter Notebook, Latex
Cloud & DevOps AWS, GCP, FastAPI, Streamlit

Experience

Aviratha Digital Labs Pvt. Ltd. — **Feb 2025 – Sep 2025**
AI/ML Intern — Bengaluru, India

- Built and deployed ML models with Python and TensorFlow, improving efficiency and cutting inference time by 25%.
- Partnered with cross-functional teams to integrate AI solutions and advanced research in computer vision and NLP with approaches for leaf disease prediction and classification.

Indian Institute of Science (IISc) — **Nov 2025 – Present**
Research Intern — Bengaluru, India

- Developing predictive models (Python, Pandas) to optimize last-mile connectivity, cutting passenger wait times by analyzing bus datasets.
- Integrating H3/OSM spatial-temporal data and collaborated with urban planners to recommend bus stop pairings, boosting transport efficiency.

Publications

The Impact of Artificial Intelligence on Side-Channel Security in the Quantum Transition — **Jan 2026**
International Conference on Secure Data Science and Applications (ICSDSA 2026) — ***[Accepted]***

KMNIST: A Large-Scale Dataset and Deep Learning Approach for Kannada Handwritten Character Recognition — **Jan 2026**
IAENG International Journal of Computer Science (IAENG-IJCS) — ***[Communicated]***

Automated Diabetic Retinopathy Grading Using EfficientNet-B0 with Grad-CAM — **Dec 2025**
International Conference on Computing for Sustainability and Intelligent Future (COMP SIF 2026) — ***[Communicated]***

Architects of Stability in a Noisy Quantum World: Learning to Outsmart Noise with Neural Alchemy — **Nov 2025**
9th International Conference on Innovative Computing and Communication (ICICC 2026) — ***[Accepted]***

AI-Driven Brain Tumor Segmentation: Innovations, Constraints and the Path Toward Clinical Adoption — **Mar 2025**
International Conference on Computing for Sustainability and Intelligent Future (COMP SIF 2025) — ***[Link]***

Poster Submission — NVIDIA GTC 2026 — **Oct 2025**

Projects

Quantum NLP Implementation with lambeq **[Link]**

- Implements an end-to-end Quantum Natural Language Processing (QNLP) pipeline using the **lambeq** toolkit, translating linguistic structures into quantum circuits via the DisCoCat framework to solve NLP tasks.

Catalyst — AI-Powered Placement Management Platform **[Link]**

- Developed a full-stack placement platform using React and Node.js, integrating Groq API (Llama 3.3 70B) for AI-powered interview simulation and career matching.
- Deployed the application on AWS using EC2 for backend hosting and configured REST APIs for integration with AI services.
- Designed responsive dashboards for students, TPOs, and administrators with a modern glassmorphic UI, ensuring smooth user workflows and scalable deployment-ready interfaces.

Tripnosis — AI-Powered Travel Recommendation System **[Link]**

- Built a full-stack AI travel planner using Python Flask, Bootstrap, and PostgreSQL/SQLite to generate personalized itineraries, flights, accommodations, and activity recommendations.
- Integrated real-time APIs including SerpApi and Groq API for travel data retrieval and AI-driven recommendation generation.
- Containerized and deployed the application on Google Cloud Run, enabling scalable hosting and secure environment configuration.

Hackathons & Competitions

- Participated in **IIIT Nagpur Quantum Computing Hackathon** — explored quantum algorithms for NLP and optimization tasks.
- Competed in **1M1B LLM Hackathon** — developed generative AI prototypes addressing real-world social impact challenges.
- Engaged in the **Google Solution Challenge** — proposed AI-based community solutions aligned with UN Sustainable Development Goals.
- Took part in **Flipkart GRID 6.0** — solved applied ML and system design problems in an industrial-scale setting.
- Participated in **BMSIT Epoch 2024** — campus-wide innovation hackathon emphasizing real-world AI/ML deployment.
- Competed in **BMSIT Code Red 2025** — designed efficient backend logic for AI-integrated applications.
- Participated in **BMSCE Hackaphasia** — collaborated on multimodal AI solutions and research-driven prototypes.
- Participated in **BMSIT Neuronova** — contributed to AI/ML innovation projects and interdisciplinary challenges.

Leadership & Extracurriculars

President — Brainium: AI/ML Student Club, BMSIT **Jul 2025 – Dec 2025**

- Leading initiatives to promote applied research, workshops, and student mentorship; engaging 200+ participants.
- Coordinated cross-institution collaborations and technical events focused on open-source AI innovation.

ML Associate — Google Developers Group, BMSIT **Oct 2025 – Feb 2026**

- Collaborating on developer-focused AI/ML initiatives, community learning, and applied research events under GDG programs.

Research Interests

Generative AI • Cognitive Systems • Natural Language Processing • Multimodal Learning • Interpretable AI • Sustainable AI • Explainable AI • Quantum-Inspired ML • Privacy-Preserving Computation